Math 520, F15 **Name:** Answer

Quiz 6 **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Find the general solution of the nonhomogeneous differential equation

$$y'' + 2y' + y = 2\sin t.$$

Solution: step 1. Find the general solution y_c of the corresponding homogeneous equation $y''+2y'+y=0$

characteristic equation: $r^2+2r+1=0$

$\Rightarrow (r+1)^2=0 \Rightarrow r_1=r_2=-1$

$\Rightarrow y_c(t) = c_1e^{-t}+c_2te^{-t}$

step 2: We set the form of a particular solution of the equation as

$Y(t) = A\cos t + B\sin t$

$\Rightarrow Y'(t) = -A\sin t + B\cos t,\ Y''(t) = -A\cos t - B\sin t$

Thus $Y''(t)+2Y'(t)+Y(t) = (A\cos t - B\sin t)+2(-A\sin t+B\cos t$

$+(A\cos t + B\sin t)$

$= \underline{2B\cos t - 2A\sin t} \equiv \underline{2\sin t}$

$\Rightarrow \begin{cases} -2A=2 \\ 2B=0 \end{cases} \Rightarrow A=-1,\ B=0 \Rightarrow Y(t)=-\cos t$

So the general solution of the given nonhomogeneous equation is

$y(t) = c_1e^{-t}+c_2te^{-t}-\cos t$

Problem 2. (4 points) Determine an appropriate form for the particular solution of each of the following nonhomogeneous differential equations:

(a) $y'' + 2y' + y = te^{2t} + t^2$

$Y(t) = (A+Bt)e^{2t} + (C+Dt+Et^2)$

(b) $y'' + 2y' + y = 2e^{-t}$

$Y(t) = At^{s}e^{-t} = At^2e^{-t}$

$s=2$